

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

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Abstract

001 Long-horizon video generation is evaluated with whole-
002 frame metrics that reward motion and temporal consis-
003 tency. For fixed-camera nature scenes this creates an
004 ambiguity: motion of water, fire, smoke, or rain is de-
005 sirable, whereas motion of the background is an error.
006 A system can therefore score well on motion while its
007 scene drifts, or on consistency while its flow stagnates.
008 We introduce SNF-Bench, an evaluation framework for
009 long-horizon fixed-camera generation that partitions
010 each scene into static support and dynamic flow and
011 reports static fidelity, flow persistence with absolute
012 magnitude, and drift leakage separately, never as one
013 score. Each factor is validated mechanistically rather
014 than by correlation with preference: we inject global
015 translation, rotation, and scale drift and progressive
016 motion attenuation at known severity into real gener-
017 ations, and require each factor to respond in its stated
018 direction and to remain selective against corruptions it
019 does not target. Auditing seven publicly released long-
020 horizon generators in their intended configurations, we
021 find that whole-frame motion and static-region drift
022 induce near-opposite orderings of the same outputs,
023 and that the controlled drift our static-fidelity fac-
024 tors detect increases a widely used whole-frame mo-
025 tion score. SNF-Bench measures where motion occurs
026 and whether it persists; it does not measure physical
027 realism.

028 1. Introduction

029 Autoregressive and streaming video diffusion models
030 now extend generation from seconds to minute-scale
031 horizons. Evaluation has not kept pace: small tempo-
032 ral errors invisible in short clips accumulate over long
033 rollouts into background displacement, colour drift,
034 or motion decay, and whole-frame metrics summarise
035 these qualitatively different failures into one score.

The problem is sharpest in fixed-camera nature
scenes. In a static-camera video of a river, the wa-
ter should move while the bank, rocks, and surround-
ing landscape stay fixed. The same separation gov-
erns waterfalls, ocean waves, rainfall, fire, smoke, and
wind-driven vegetation: some regions form static sup-
port that must remain structurally fixed, while others
carry dynamic flow that must persist. Whole-frame
motion measures reward optical flow regardless of its
origin, crediting background drift as readily as genuine
fluid motion, while temporal-consistency measures can
favour videos whose stability comes from suppressing
the flow they should sustain. Two opposite failures—
static drift and flow decay—are thereby conflated, and
increasingly so at long horizons, where accumulated
background motion can dominate a sequence’s appar-
ent dynamics. Figure 1 shows both failures in released
generations of a single prompt.

We introduce SNF-Bench (Static-Natural Flow), an
evaluation framework built around this decomposition,
evaluated at 60 and 120s horizons with shorter and
longer tiers retained as diagnostics. Measurement is
organised into three factors. Static fidelity captures
geometric displacement and residual optical flow inside
nominally stationary regions. Flow persistence reports
the retention of dynamic-region motion together with
its absolute compensated magnitude, so a retention
ratio cannot appear strong merely because motion is
uniformly weak. Drift leakage relates static-region mo-
tion and global-motion compensation to the measured
dynamic-region flow.

A benchmark is only as trustworthy as its measure-
ments, so we validate the protocol independently of
any model comparison. Controlled perturbations in-
ject global translation, rotation, and scale drift, pro-
gressive motion attenuation, late freezing, photometric
drift, and temporal repetition into real fixed-camera
generations at known severity, and we require each fac-
tor to respond in the direction its definition states and
to remain selective against corruptions it does not tar-

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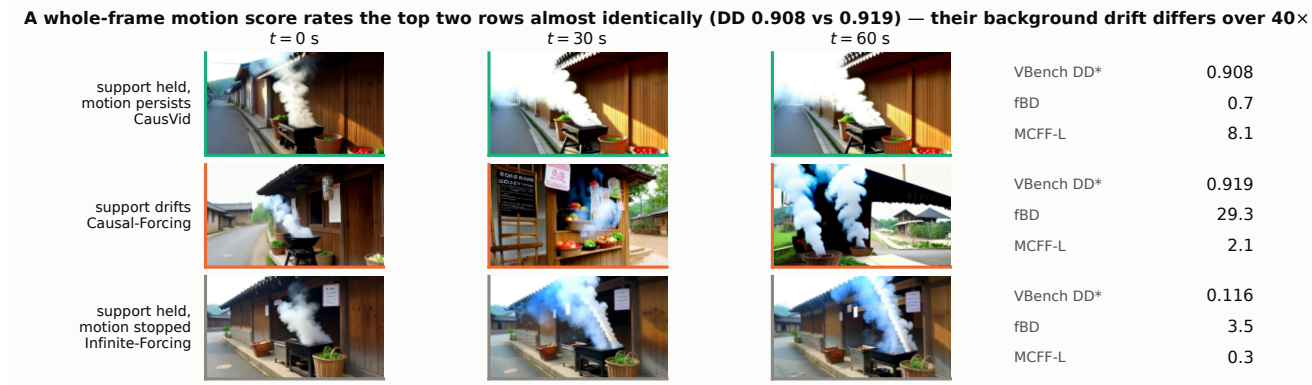


Figure 1. Why a fixed-camera benchmark needs a spatial decomposition. Three released 60s generations of one prompt. A whole-frame motion score rates the first two rows almost identically, yet only the first holds its scene in place; their static-region drift differs by over an order of magnitude. The third row shows the opposite failure, which that score does detect.

get. Because a parameter expressed in degrees is not commensurate with one expressed in pixels, geometric severity is specified as the mean displacement a corruption induces, which makes the families directly comparable. We additionally recompute every measurement under systematic erosion and dilation of the region partition.

We then audit seven publicly released long-horizon generators in their intended configurations. The audit exposes cases where conventional metrics and SNF-Bench interpret the same outputs differently: high whole-frame motion coexisting with substantial static-region displacement, and high stability coexisting with decaying flow. We do not argue that existing metrics are incorrect. They answer a whole-frame question accurately; that question does not encode the spatial distinction this setting requires. Nor does SNF-Bench measure physical realism—a sequence may be drift-free and persistent yet physically implausible. The contribution is narrower and complementary: making the spatial origin and temporal persistence of motion explicit.

Our contributions are:

1. SNF-Bench, an evaluation framework for long-horizon fixed-camera generation organised around a static-support/dynamic-flow partition.
2. A measurement protocol that reports static fidelity, flow persistence with absolute magnitude, and drift leakage separately, and never as a single score.
3. Mechanistic validation of every reported factor under corruptions of known type and severity, including calibration of the drift diagnostics and robustness of the region partition.
4. A public-model audit showing that this decomposi-

tion materially changes the interpretation and ordering of long-horizon behaviour relative to whole-frame metrics.

2. Related Work

Video generation evaluation. General-purpose benchmarks assess complementary aspects of visual quality, temporal consistency, motion, and semantic alignment; VBenCh [2], for example, decomposes quality into dimensions including Background Consistency, Motion Smoothness, Temporal Flickering, and Dynamic Degree. These suites enable broad comparison across heterogeneous generators, but their measurements are primarily whole-frame or sequence-level. In fixed-camera nature-flow scenes, whole-frame aggregation cannot reveal whether measured motion originates in intended dynamic regions or in displacement of nominally static structure. SNF-Bench complements such suites by making the spatial origin of motion explicit; it replaces none of them.

Long-horizon autoregressive generation. Autoregressive and streaming generators extend synthesis beyond short bidirectional clips by producing frames or temporal chunks causally, including CausVid [10], Self-Forcing [1], Infinite-Forcing [3], Rolling-Forcing [5], Reward-Forcing [6], LongLive [8], and Causal-Forcing [11]. These systems differ substantially in temporal block size, training objective, memory and cache policy, motion prior, and horizon-extension mechanism [9]. Their evaluation, however, typically relies on generic video metrics or method-specific qualitative analysis. The deployment diversity motivates SNF-Bench’s two-setting audit—native capability un-

142	der each released configuration, and a separately re-	191
143	ported matched stress test—while the evaluation gap	192
144	motivates its method-agnostic question: does static	193
145	support remain fixed while dynamic flow persists?	194
146	Image dynamics and nature-flow animation. Work on	195
147	animating still images has long modeled natural motion	196
148	explicitly: controllable fluid animation separates static	
149	image structure from specified fluid motion [7], and	
150	Generative Image Dynamics models long-term scene	
151	motion through learned trajectory representations [4].	
152	These works affirm that static support and dynamic	
153	content are distinct modeling targets—the same dis-	
154	tinction SNF-Bench elevates to an evaluation principle	
155	for modern long-horizon generators.	
156	Image-to-video faithfulness. I2V generation adds a	
157	reference signal: the source image specifies the ini-	
158	tial scene. Existing I2V evaluation commonly mea-	
159	sures source or subject consistency alongside temporal	
160	quality, but high source similarity does not determine	
161	whether motion remains confined to regions expected	
162	to move, nor whether stability was achieved by sup-	
163	pressing motion. SNF-I2V applies the same static-	
164	dynamic decomposition as SNF-T2V, adding source-	
165	referenced static-region fidelity as one complementary	
166	factor.	
167	3. SNF-Bench Design	
168	SNF-Bench evaluates long-horizon generation under a	
169	deliberately constrained setting: a fixed camera observ-	
170	ing a scene containing both stationary structure and	
171	persistent natural motion. This isolates two failure	
172	modes that are difficult to separate in unconstrained	
173	video—static drift, where nominally stationary struc-	
174	ture moves over time, and flow decay, where intended	
175	dynamics weaken or disappear. Every sequence is	
176	treated as a composition of two spatial roles, Ω_{static}	
177	for regions expected to remain geometrically station-	
178	ary and Ω_{flow} for regions expected to exhibit sustained	
179	motion. The partition defines an evaluation role, not a	
180	semantic ontology: a riverbank is static support while	
181	the river surface is flow; buildings are static support	
182	in a rainfall scene while the falling rain is flow. SNF-	
183	Bench does not require dynamics inside Ω_{flow} to follow	
184	any physical model. It measures whether motion stays	
185	localised to the appropriate regions and persists over	
186	time without contaminating static support.	
187	Every system receives an identical prompt per	
188	benchmark item, and no prompt is rewritten for a par-	
189	ticular method. The main paper evaluates the text-	
190	conditioned setting, where the opening of the sequence	
	establishes the scene’s static support and evaluation	191
	asks whether that support survives the rollout while	192
	flow remains active. An exploratory image-conditioned	193
	analysis, in which the source image specifies static sup-	194
	port externally, is reported in the supplementary ma-	195
	terial and forms no part of the claims below.	196
	3.1. Scene Categories and Horizons	197
	Prompts are stratified across representative fixed-	198
	camera natural processes—channel water, ocean waves,	199
	precipitation, fire and smoke, lava, and wind-borne	200
	motion—with the category set fixed before the final	201
	sweep. The strata are not equally populated: chan-	202
	nel water accounts for 61% of the principal prompt	203
	set, which is why every reported aggregate is macro-	204
	averaged over the 6 categories rather than averaged	205
	over prompts, so that no single medium determines a	206
	headline value. Complete per-category counts are given	207
	in the supplementary material.	208
	Four temporal regimes are used: 5s as an initial-	209
	isation check, 60 and 120s as the reported horizons,	210
	and 240s as a diagnostic stress test. Principal conclu-	211
	sions rest on the 60s tier, which carries 23 prompts;	212
	the 120s tier is reported as a supporting horizon and	213
	its sample does not support method-level separation on	214
	its own. A system that cannot reach a horizon under	215
	its released configuration is reported as unsupported	216
	rather than silently modified.	217
	3.2. Region Partition	218
	The partition is obtained automatically and identically	219
	for every sequence. Mean optical-flow magnitude over	220
	the leading 12% of sampled frame pairs is thresholded	221
	by Otsu’s method into a dynamic region and its com-	222
	plement; the static region is then eroded and a 4%	223
	frame border discarded, which together form an ig-	224
	nored transition band so that boundary pixels—where	225
	the partition is least certain and flow least reliable—	226
	contribute to neither region.	227
	Restricting the estimate to an early window is what	228
	prevents the measurement from becoming circular. De-	229
	riiving a partition from a sequence’s entire motion field	230
	would allow accumulated background drift to be read	231
	as evidence that the affected region is dynamic; at the	232
	60s horizon the partition is instead fixed from the first	233
	7.1s, before long-horizon drift has accumulated, and	234
	the metrics are evaluated over the remainder. The	235
	partition therefore reflects each system’s spatial lay-	236
	out, which is unavoidable when every system authors	237
	its own scene, but it is not defined by the temporal	238
	failure being measured.	239
	Two limitations follow, and both bound the conclu-	240
	sions drawn later. The partition is automatic rather	241

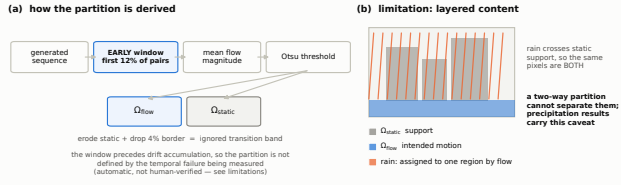


Figure 2. Region-partition protocol. (a) The partition is derived automatically from early-window flow, with an ignored transition band at the boundary. (b) A two-way partition cannot represent layered content such as rain crossing static support.

than annotated, so a system that drifts within the first 12% of its rollout can influence its own partition; we therefore treat partitions as measured quantities and recompute every result under systematic erosion and dilation of the boundary (Sec. 5). And a two-way partition cannot represent layered content: rain and snow transit static support, so those pixels are assigned to one region by flow magnitude alone and static-fidelity values in precipitation scenes carry an additional unmodelled contribution. Derivation details, the sensitivity study, and the annotation protocol we regard as the natural strengthening of this component are given in the supplementary material.

3.3. Benchmark Manifest

SNF-Bench is defined by a versioned manifest fixed before the final sweep, recording per item: identifier and category, exact prompt, target horizons, resolution and frame-rate metadata, and provenance. After freezing, items are never removed because a system performs poorly; corrections require a new benchmark version. A failure taxonomy—unsupported horizon, out-of-memory, runtime crash, numerical corruption, invalid output—is likewise fixed in advance, and failures are reported alongside successful generations rather than silently excluded. Coverage is counted from valid per-video measurements rather than from the presence of a results file, so a run that terminated early is reported as partial.

4. Drift–Motion Evaluation Protocol

SNF-Bench evaluates fixed-camera generation along three complementary factors: Static Fidelity, Flow Persistence, and Drift Leakage. No individual score is sufficient: low background motion can be achieved by freezing the entire video, and high foreground motion can be produced by drifting the entire scene. We therefore always report static-region stability jointly with both the relative persistence and absolute magnitude of dynamic-region motion.

Let I_t denote frame t of a T -frame sequence, Ω_{static} and Ω_{flow} the fixed region masks of Sec. 3.2, and $\mathbf{u}_t(\mathbf{x})$ the optical flow from I_t to I_{t+1} at pixel $\mathbf{x}$. Each sequence defines an early window $\mathcal{E}$ and a late window $\mathcal{L}$ of equal duration; $\mathcal{E}$ begins after a short warm-up interval so that initial generation transients do not define the motion reference. Window extents are fixed in the benchmark specification.

4.1. Static Fidelity

Feature-aligned Background Drift (fBD). We extract repeatable local features inside Ω_{static} from the first frame and match them against each frame I_t in the late window, applying robust geometric filtering to reject outlier correspondences. For the resulting matched pairs $\mathcal{M}_t$,

$$D_{\text{feat}}(t) = \text{median}_{(p,q) \in \mathcal{M}_t} \|p - q\|_2,$$

$$\text{fBD} = \frac{100}{d} \cdot \frac{1}{|\mathcal{L}|} \sum_{t \in \mathcal{L}} D_{\text{feat}}(t) \quad (\downarrow), \quad (1)$$

where d is the frame diagonal, so fBD is expressed as a percentage of image size. fBD captures accumulated geometric displacement of persistent structure and is intentionally insensitive to fine-scale flicker that moves no structure.

Normalized Background Flow (NBF). Complementing accumulated drift, NBF measures instantaneous motion energy inside the static mask:

$$\text{NBF} = \frac{10^3}{W \Delta t} \cdot \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{|\Omega_{\text{static}}|} \sum_{\mathbf{x} \in \Omega_{\text{static}}} \|\mathbf{u}_t(\mathbf{x})\|_2 \quad (\downarrow), \quad (2)$$

normalized by frame width W for resolution invariance and by the inter-frame interval Δt for frame-rate invariance, giving units of 10^{-3} frame widths per second. Here Δt is the interval of the sampled pairs actually used for flow estimation: methods are evaluated at their native frame rates, which differ across the roster, and the evaluation pipeline subsamples each clip to a common temporal rate before measurement. Expressing NBF per unit time rather than per frame makes the quantity independent of both the native rate and the sampling policy, so the measure remains comparable if either changes.

NBF is a normalized magnitude, not a ratio: we deliberately do not divide by whole-frame flow, since that would couple background leakage to foreground behavior and inflate the measure for methods whose dynamic motion decays.

4.2. Flow Persistence

Global-motion compensation. Coherent scene displacement is removed before dynamic-region motion is evaluated. A single translation vector is insufficient: under rotation or zoom the static-region flow field is not constant, and its component-wise median is near zero for a centered rotation, so a translation-only estimate would leave exactly the corruptions Sec. 5 injects uncompensated. We therefore estimate a robust global similarity transform $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$ from static-region correspondences $\mathbf{x} \mapsto \mathbf{x} + \mathbf{u}_t(\mathbf{x})$ under RANSAC, define the induced global displacement field $\mathbf{d}_t(\mathbf{x}) = T_t(\mathbf{x}) - \mathbf{x}$, and compensate pointwise:

$$\tilde{\mathbf{u}}_t(\mathbf{x}) = \mathbf{u}_t(\mathbf{x}) - \mathbf{d}_t(\mathbf{x}). \quad (3)$$

We restrict the model to a similarity rather than an unrestricted affine transform because affine shear can absorb legitimate local deformation and over-correct. Where correspondences are insufficient or the robust fit fails we fall back to median translation and record that fallback per sequence.

Motion-Compensated Foreground Flow (MCFF). For a temporal window $\mathcal{W}$,

$$\text{MCFF}(\mathcal{W}) = \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \frac{1}{|\Omega_{\text{flow}}|} \sum_{\mathbf{x} \in \Omega_{\text{flow}}} \|\tilde{\mathbf{u}}_t(\mathbf{x})\|_2 \quad (\uparrow). \quad (4)$$

We report $\text{MCFF}(\mathcal{E})$ and $\text{MCFF}(\mathcal{L})$, so a method with uniformly negligible motion cannot appear strong through relative measures alone.

Flow Persistence (FP).

$$\text{FP} = \min\left(\frac{\text{MCFF}(\mathcal{L})}{\text{MCFF}(\mathcal{E}) + \epsilon}, c\right) \quad (\uparrow), \quad (5)$$

clipped at a fixed constant c to prevent unstable ratios when early motion is small. FP near one indicates retained motion; substantially smaller values indicate decay; values above one are not automatically better. FP is never reported without its MCFF magnitudes.

4.3. Drift Leakage and Attenuation

Whole-frame metrics cannot determine whether apparent dynamic-region motion arises from intended flow or from displacement of the whole scene. We report two diagnostics, since neither subsumes the other. Let F_{static} denote mean flow magnitude over Ω_{static} , and $F_{\text{raw}}, F_{\text{comp}}$ the mean raw and compensated flow magnitude over Ω_{flow} , all in the late window; the late window is used because drift accumulates over the rollout.

Drift Leakage Ratio (DLR).

$$\text{DLR} = \frac{F_{\text{static}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow) \quad (6)$$

locates the sequence’s motion energy. It is not a fraction and is deliberately unbounded: $\text{DLR} > 1$ indicates that static support moves more than the region intended to move, a regime that occurs in practice.

Drift Attenuation Ratio (DAR).

$$\text{DAR} = 1 - \frac{F_{\text{comp}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow) \quad (7)$$

measures the signed relative change in dynamic-region flow after compensation—how much apparent motion a global displacement model accounts for. Reported values are clipped to $[0, 1]$ while the signed quantity is retained for validation; negative values, which arise where local flow opposes the estimated global field, are reported rather than suppressed.

Interpretation. Neither quantity is a causal decomposition of motion into drift and non-drift components, and we do not describe them as one. Read together they are more informative than either alone: high DLR with moderate DAR indicates large static-region motion energy that a rigid global model does not explain, implicating non-rigid instability such as warping rather than uniform scene translation.

4.4. Joint Interpretation

No factor is read alone. Low static drift together with low compensated motion indicates an over-stabilised, nearly frozen sequence; high compensated motion together with high static-region flow indicates motion contaminated by displacement; high retention with negligible magnitude indicates persistent but trivial motion. The desired regime is low static drift, non-trivial compensated motion, and high retention. SNF-Bench therefore reports no single scalar score: collapsing the factors would recreate the ambiguity it exists to expose.

Implementation convention. Videos are normalised to a common evaluation resolution and temporally sub-sampled to a common flow-estimation rate; no frame interpolation is performed, since synthesising intermediate frames would alter the flow field being measured. Every estimator, threshold, window extent, and constant is fixed globally before the sweep and applied unchanged to every system; the full list, and the per-sequence frame rate, resolution and compensation mode released alongside the scores, are given in the supplementary material.

5. Controlled Validation

Before SNF-Bench is used to compare generators, we validate that its measurements respond to the failure modes they target. A benchmark is not justified merely because its metrics produce different rankings; each metric must change selectively, monotonically, and in the expected direction under known corruptions. We therefore construct controlled variants of fixed-camera reference clips with known corruption type and severity. Validation is performed independently of the audit in Sec. 6: perturbation strengths, masks, temporal windows, and metric hyperparameters are fixed before the final baseline results are inspected.

5.1. Perturbations and Criteria

Given a reference sequence we generate variants that isolate individual components of the decomposition. Geometric drift applies a temporally increasing global transformation—translation, rotation, or scale—simulating camera-like background displacement that accumulates as long-horizon error does. Motion attenuation progressively blends dynamic-region content toward an early reference, and late freezing replaces all content after a chosen fraction of the sequence, both leaving static support untouched. Photometric drift varies brightness and colour temperature while preserving geometry and motion, bounding the residual sensitivity of flow-based quantities. Temporal repetition replaces late dynamic content with a short repeated cycle. Partition perturbation erodes and dilates the region boundary rather than the video.

Geometric severity is specified as the mean pixel displacement a corruption induces over the frame, and each family’s own parameter is derived from it using the frame geometry. This is not a presentational choice: a parameter expressed in degrees is not commensurate with one expressed in pixels, and at the evaluation resolution one degree of rotation induces only a few pixels of mean displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude.

A factor is admitted to the audit only if its response direction matches its definition, that response is monotonic in severity, unrelated corruptions produce substantially smaller changes, and conclusions survive perturbation of the region partition. Factors failing these checks are revised or reported as supplementary diagnostics rather than headline quantities. Complete perturbation levels, the calibration procedure for the drift diagnostics, and the full response matrix are given in the supplementary material.

perturbation	clips	fBD	NBF	MCFF-L	FP	DLR	DAR	VB-DD
photometric	7	+0.30	+0.90	+0.70	+0.60	-0.90	-0.90	+0.60
mask radius	7	+0.90	+1.00	-0.50	-0.90	+1.00	-0.70	—
translation	5	+1.00	+1.00	-0.90	-0.10	+1.00	-0.70	+0.70
rotation	5	+0.90	+0.00	-0.70	-0.30	-0.90	+0.60	+0.90
scale	5	+1.00	+1.00	-0.90	-0.60	+0.10	+0.90	+0.30
attenuation	6	-0.10	-1.00	+0.00	-0.10	+1.00	-1.00	-1.00
freeze	6	-1.00	-1.00	-1.00	-0.40	-0.40	+0.20	-1.00
repetition	6	-0.20	-0.20	+0.20	+0.20	-0.40	-0.40	-0.40

Table 1. Mechanistic validation. Spearman correlation between injected severity and each factor’s response, over controlled perturbations of real fixed-camera clips. Signs are fixed in advance by each definition. mask radius perturbs the region partition rather than the video, so near-zero entries indicate the desired insensitivity. Cells where a factor responds weakly or non-monotonically are reported rather than omitted.

5.2. Validation Results

Figure 3 and Table 1 report the outcome. Injected translation produces the expected monotone increase in both static-fidelity factors, and progressive attenuation of dynamic-region content drives MCFF-L and FP down while leaving the static factors comparatively unchanged—the selectivity the decomposition requires. The comparison that motivates the benchmark appears in the same experiment: the identical injected background drift that our static-fidelity factors register also raises VBench Dynamic Degree, so a whole-frame motion score improves on precisely the corruption a fixed-camera evaluation should penalise.

Two results are reported because they qualify the framework rather than support it. fBD responds strongly and monotonically to translation but behaves erratically under pure rotation, where feature correspondences degrade; NBF carries the static-fidelity signal in that regime, which is why the two are reported jointly and why neither is used alone. Reported alongside them is the scope boundary of Sec. 5.1: looping dynamic content sustains both persistence and magnitude, so SNF-Bench cannot distinguish genuine progression from repetition, and does not claim to.

6. Public-Model Audit

We apply SNF-Bench to publicly released long-horizon generators. The goal is not a universal leaderboard but a determination of how deployed systems occupy the static-fidelity/flow-persistence space when used as their authors intended.

Systems and setting. The audit covers 7 publicly released autoregressive or streaming generators with accessible checkpoints and reproducible inference code,

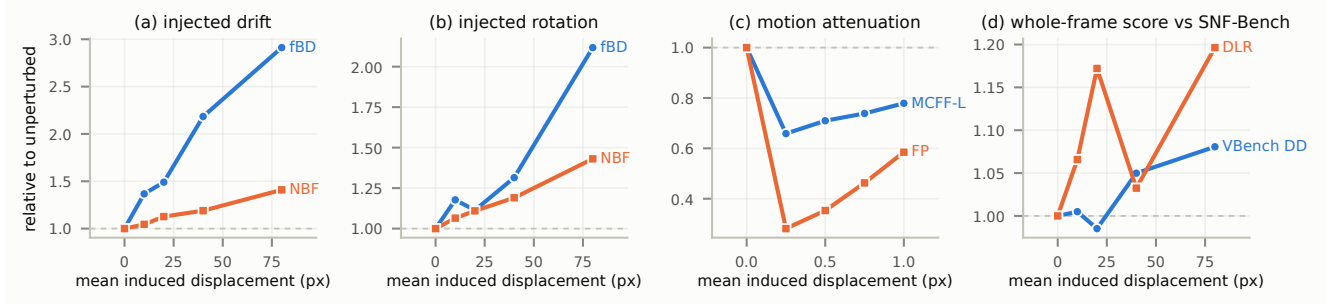


Figure 3. Mechanistic validation. Each panel injects a corruption of known kind and severity into real fixed-camera clips and plots each factor relative to its unperturbed value. Panel (f) puts VBench Dynamic Degree through the identical corruption as the SNF factors: the score rises with injected background drift. The perturbation is synthetic, so the comparison implicates no published method.

frozen as a set before the sweep. Unpublished and internal systems are excluded by construction, as are systems whose outputs cannot be independently reproduced. Each is run in the deployment regime its release intends, answering a capability question: what behaviour does the released system exhibit when used as documented? Differences in temporal block size, denoising schedule, and memory policy are preserved deliberately—they are part of the released system, not nuisance variables—so the results compare deployed systems rather than isolated checkpoints under matched compute. The audited systems share a deployment envelope: all generate 832×480 at 16 fps and all reach the 240s horizon without modification to their released temporal policy. A system unable to reach a horizon without altering a mechanism central to its released procedure would be marked unsupported there; we never substitute a block size, cache policy, or schedule to populate a table entry.

A secondary analysis evaluates checkpoints under a common long-horizon wrapper, measuring sensitivity to deployment regime rather than capability. Those scores are never used to redefine native behaviour or attributed to the published system, and are reported in the supplementary material.

Controls and statistics. Within each benchmark item every system receives identical conditioning, and the seed set is fixed before the sweep. Seed control guarantees reproducibility within a system; because implementations consume randomness differently, identical seeds do not induce matched noise draws across systems, so cross-system comparison rests on the paired-item design rather than on paired noise. Generated videos are retained in native form, and the normalisation of Sec. 4 is applied unchanged to every system.

Scores are computed per item and macro-averaged

across categories. Uncertainty is estimated by percentile bootstrap over items, and we report 95% intervals. Exact ordinal ranks are avoided where uncertainty does not support them: systems whose intervals overlap substantially are not described as separated. Operational limits—unsupported horizons, failed generations, memory exhaustion, invalid outputs—are reported as rates against the frozen manifest and never removed after inspection.

To test whether the decomposition changes interpretation, the same outputs are additionally scored with established whole-frame metrics and we compare the orderings they induce. The focus is interpretation change, not a claim that existing metrics are incorrect.

7. Results and Interpretation Changes

Unless stated otherwise, comparisons refer to systems in their released configurations; deployment-sensitivity results are reported separately in the supplementary material and never redefine a system’s native behaviour.

7.1. Native-Capability Audit

Table 2 summarizes SNF-Bench measurements under released inference configurations at the two principal horizons, with all four reported in the supplementary material. The audited systems happen to share a deployment envelope—all seven generate 832×480 at 16 fps and all reach the 240s horizon without modification to their released temporal policy—so no per-system configuration table is needed; where a system could not reach a horizon, the entry would be marked unsupported rather than populated with altered settings.

Motion persistence. Dynamic-region behavior is evaluated at the joint operating point (MCFF($\mathcal{L}$), FP),

Method	fBD↓	NBF↓	MCFF-L↑	FP↑	DLR↓
60s horizon ($n = 23$)					
CausVid	8.08 ± 3.15	9.27 ± 4.14	1.96 ± 1.33	0.67 ± 0.18	0.45 ± 0.13
Self-Forcing	11.4 ± 3.63	12.7 ± 3.97	2.54 ± 1.40	0.44 ± 0.22	0.50 ± 0.19
Infinite-Forcing	7.75 ± 3.84	3.65 ± 0.88	0.29 ± 0.09	0.63 ± 0.22	0.55 ± 0.11
Rolling-Forcing	16.8 ± 3.69	11.2 ± 2.70	1.57 ± 0.88	0.75 ± 0.29	0.54 ± 0.12
Reward-Forcing	14.8 ± 3.71	8.51 ± 2.79	0.80 ± 0.30	0.49 ± 0.18	0.60 ± 0.13
LongLive	15.9 ± 2.89	13.9 ± 4.76	2.41 ± 1.98	0.75 ± 0.28	0.59 ± 0.13
Causal-Forcing	22.4 ± 2.26	87.6 ± 18.6	7.21 ± 4.17	0.83 ± 0.31	0.94 ± 0.20
120s horizon ($n = 6$)					
CausVid	6.15 ± 5.35	3.42 ± 3.27	0.52 ± 0.54	1.06 ± 0.40	0.41 ± 0.07
Self-Forcing	21.2 ± 5.69	6.80 ± 3.50	0.82 ± 0.54	0.85 ± 0.63	0.75 ± 0.48
Infinite-Forcing	16.1 ± 5.56	2.60 ± 0.92	0.32 ± 0.33	0.64 ± 0.24	0.57 ± 0.20
Rolling-Forcing	23.8 ± 3.91	10.6 ± 5.27	0.77 ± 0.59	0.38 ± 0.28	0.82 ± 0.18
Reward-Forcing	20.0 ± 5.01	11.6 ± 5.23	0.69 ± 0.42	0.39 ± 0.21	0.70 ± 0.17
LongLive	22.5 ± 6.33	9.66 ± 3.99	1.19 ± 0.99	0.81 ± 0.49	0.84 ± 0.37
Causal-Forcing	21.8 ± 1.41	130 ± 55.9	10.7 ± 12.0	0.90 ± 0.39	0.81 ± 0.21

Table 2. SNF-Bench audit, T2V track, native configuration. All systems are public external models run under their own intended configuration. Our own systems are excluded by construction. DAR is reported clipped to $[0, 1]$; signed values are released. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel. Values are prompt-level means $\pm$ half the width of a percentile bootstrap 95% interval (10k resamples, fixed seed). MCFF-E and DAR appear in the supplementary table.

never FP alone: sustained flow requires non-trivial late motion and high retention. Reporting MCFF-E beside MCFF-L makes the degenerate case visible—a system with negligible motion throughout cannot reach a high FP without also displaying a small MCFF-E.

Static–dynamic operating regimes. Figure 4 places each system in the static-drift–surviving-flow plane, exposing qualitatively different regimes—low-drift/low-motion, high-motion/high-drift, and low-drift/non-trivial-motion—without collapsing the axes into a scalar score. The two extremes are informative in opposite directions: Infinite-Forcing attains the lowest static drift of any audited system while placing last on surviving late motion, and Causal-Forcing sits alone at the opposite corner.

7.2. Drift Leakage Changes the Interpretation of Apparent Motion

The effect is systematic rather than anecdotal. Over the audited systems at 60s, VBench Dynamic Degree correlates with SNF-Bench’s static-region flow at Spearman $\rho = +0.93$ (Table 3): the systems a generic metric reads as most dynamic are, to a close approximation, the systems whose backgrounds move most. The consequence for ordering is direct. Causal-Forcing ranks 1 of 7 on Dynamic Degree and 7 of 7 on NBF; Infinite-Forcing occupies the mirror position, ranking 1 on static fidelity and 7 on apparent motion. Read

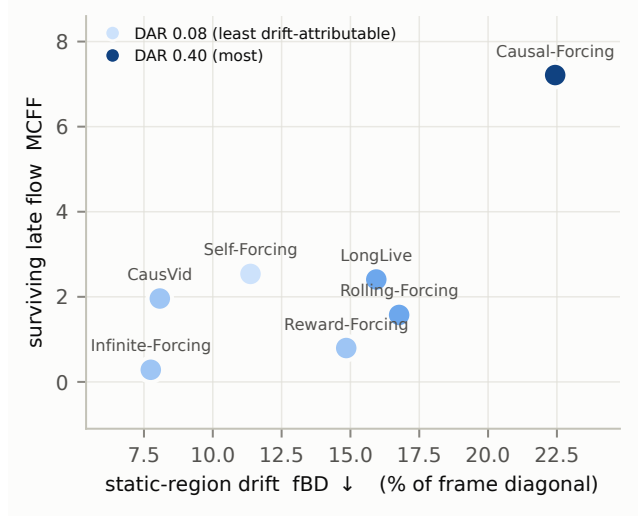


Figure 4. Operating regimes under native deployment. Colour encodes the share of dynamic-region flow a global similarity model accounts for. The view exposes trade-offs; it is not a leaderboard.

generic metric	fBD	NBF	MCFF	DLR
VB-DD	+0.75	+0.93	+0.89	+0.36
VB-bg	-0.64	-0.71	-0.75	-0.50
VB-smooth	-0.96	-0.82	-0.61	-0.64
VB-flick	-0.96	-0.82	-0.61	-0.64

Table 3. Rank disagreement at 60s. Spearman correlation between generic whole-frame metrics and SNF-Bench factors over method-level means. A positive correlation between apparent motion and background flow is the effect the benchmark isolates.

jointly with its magnitudes, Causal-Forcing’s DLR of 0.94 indicates static-region motion energy almost equal to that of the region intended to move, while its DAR of 0.40 indicates that a rigid global model accounts for only part of it—the remainder is non-rigid instability, which a translation-only account would have misdescribed.

The supplementary material records every interpretation change meeting three conditions fixed in advance: both measurements are computed on the same outputs, the SNF factor involved passed the validation of Sec. 5, and the rank difference exceeds a fixed threshold rather than reflecting a small ordinal shift. The rule is applied to every audited system, so no qualifying case can be selectively omitted. The analysis is complementary: whole-frame metrics measure what they were designed to measure, and SNF-Bench exposes where that quantity spatially originates.

7.3. Category and Horizon Dependence

Aggregates can conceal scene dependence, so all headline numbers are macro-averaged over the 6 flow-medium categories of Sec. 3 rather than averaged over prompts directly. This matters: the 60s prompt set is 61% channel water, so a prompt-level mean is substantially a measurement of one medium. Complete per-category values appear in the supplementary material, together with the category balance of every track and horizon.

Horizon dependence is read from the per-horizon panels of Table 2. A useful long-horizon generator should not merely score well at one duration but remain comparatively stable as the horizon grows. The 5s tier serves as an initialization sanity check; the 240s tier is exploratory and determines no principal conclusion, but it is reported because the ordering it induces is itself evidence about metric behavior under extended rollout.

8. Limitations and Conclusion

Scope. SNF-Bench is intentionally narrow: long-horizon fixed-camera generation, where static support and dynamic flow can be meaningfully separated. It does not rank general-purpose generation across moving-camera cinematography, articulated action, narrative scene change, or interactive simulation, and its results should be read as evidence about this controlled regime.

Persistence is not realism. The benchmark measures whether motion occurs in the expected regions, remains non-trivial over time, and is uncontaminated by background displacement. It does not determine whether water, smoke, or fire obey physically correct dynamics: a sequence can be drift-free and persistent while remaining implausible, and persistence cannot distinguish genuine progression from a repeated cycle. Perceptual evaluation and physics-aware diagnostics remain complementary rather than superseded.

The partition and the estimators are approximate. The partition is derived automatically from early-window flow rather than from annotation, so a system that drifts within the opening of its rollout can influence its own partition, and layered content such as rain crossing static support cannot be represented by a two-way split. The transition band and the erosion-dilation study bound this uncertainty without removing it, and conclusions unstable under partition perturbation are not reported as headline claims. All measurements likewise inherit the failure modes of optical-flow and fea-

ture estimators: controlled validation establishes their behaviour over the benchmark’s operating range, and they should be read as calibrated diagnostics rather than exact physical decompositions. Human-verified first-frame annotation is the natural strengthening of this component and the clearest direction for a subsequent version.

Conclusion. We introduced SNF-Bench, an evaluation framework for long-horizon fixed-camera video generation built on one distinction: some regions of a scene must hold still, and others must not stop moving. The framework measures static fidelity, flow persistence together with absolute magnitude, and drift leakage; it validates every reported factor against corruptions of known type and severity before any system is compared; and it audits publicly released generators in the configurations their authors intended. Whole-frame metrics answer whether a video moves and whether it is consistent, and they answer that accurately. SNF-Bench answers what those aggregates cannot: where the motion originates, and whether it lasts. As horizons extend from seconds to minutes, that spatial accounting becomes the difference between a scene that flows and one that merely drifts.

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